

Business 0.2

Business Validation & Lean Canvas — Africa Mineral Intelligence Platform

1. One-Sentence Value Proposition

“A centralized African mineral intelligence platform that helps governments, mining companies, investors, and researchers monitor, analyze, and make decisions using real-time geospatial mining data and analytics.”

2. Single Primary Customer Segment

Primary Segment (Start Here)

Government Mining & Regulatory Agencies

Starting with:

- Ghana Minerals Commission
 - Geological Survey Authority
 - Environmental Protection Agency Ghana
-

Why This Segment First?

They:

- already need centralized mining intelligence
- control mining datasets
- have recurring operational needs
- can become anchor customers
- increase platform credibility

This avoids the classic early-stage mistake of trying to serve:

- governments
- miners
- researchers

- NGOs
- investors

all at once.

3. Monetisation / Business Model Hypotheses

Option 1 — SaaS Subscription Model (Recommended MVP)

Customers

- Mining companies
- Government agencies
- Universities
- Investors

Pricing Example

Tier	Monthly Price
Basic	\$99/month
Professional	\$499/month
Enterprise	\$2,000+/month

Option 2 — Data/API Usage Model

Charge for:

- API calls
- downloadable datasets
- geospatial analytics
- export reports

Example

- \$0.01–\$0.10 per API request

- Premium dataset licensing
-

Option 3 — Enterprise & Government Contracts

Large customized deployments:

- regulatory monitoring systems
- illegal mining tracking
- environmental dashboards
- national mineral intelligence systems

Revenue Potential

High-ticket contracts:

- \$20k–\$500k+

This may become the strongest long-term model in Africa.

4. TAM / SAM / SOM Estimation

A. Top-Down Market Estimation

TAM (Total Addressable Market)

Global mining software & mining intelligence market:

Estimated:

~\$15B–\$25B+

Includes:

- GIS systems
- exploration intelligence
- mining ERP
- analytics
- monitoring platforms

SAM (Serviceable Available Market)

African mining intelligence & monitoring segment:

Estimated:

~\$150M–\$500M

Focus:

- ECOWAS
- mining agencies
- mining operators
- mineral analytics users

This exceeds the required:

> \$10M SAM threshold

SOM (Serviceable Obtainable Market)

Realistic 3–5 year obtainable market:

Starting with:

- Ghana
- selected ECOWAS countries

Estimated achievable market:

~\$2M–\$10M ARR

through:

- government contracts
 - SaaS subscriptions
 - analytics licensing
-

B. Bottom-Up Market Estimation

Ghana Example

Potential Customers

Segment	Estimated Customers
Large mining companies	20–40
Mid-size mining firms	50–100
Government agencies	5–15
Universities/research bodies	20–40
Investors/consultancies	50+

Revenue Estimate

Assume:

- 50 organizations
- average annual contract = \$15,000

Annual Revenue

$$50 \times \$15,000$$

$$= \$750,000/\text{year (Ghana only)}$$

ECOWAS Expansion

If expanded across:

- Ghana
- Nigeria
- Guinea
- Mali
- Burkina Faso
- Côte d'Ivoire

You could realistically reach:

\$3M–\$10M ARR

without needing massive user volume.

5. Lean Canvas — Africa Mineral Intelligence Platform